ALP-ttbar Toy MC v1.9 — Preregistration: Layer Geometry at Fixed Material for the All-Pixel Three-Dimensional Vertex Fit of Displaced Light ALP → e^+e^-

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Zenodo: doi:10.5281/zenodo.24048262 · GitHub: github.com/wspaik5/ALP_ttbar_Toy_MC_v1.9_Preregistration_Layer_Geometry
SHA-256 of the frozen README.md: 9370cf0a0519d8075340655c1c181684b15557784f8a0a943b777059b6f40085

> **About this document.** This PDF reproduces the frozen preregistration text (README.md) unchanged and adds two figures made only from the geometry definitions, an analytic resolution formula and published v1.6/v1.8 numbers. No v1.9 simulation or fit had been run when it was written. **Disclaimer.** Independent personal toy study, not affiliated with the authors of arXiv:2610.38932 or arXiv:2306.08686, nor with the CMS, LHCb or ALICE Collaborations or the FCC study. Detector numbers are assumed; only qualitative trends and ratios are meaningful.

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Date frozen: 2026-11-08 (this file is committed to GitHub before any v1.9 simulation or fit is run)
Builds on: v1.8, https://doi.org/10.5281/zenodo.24034481 (preregistration https://doi.org/10.5281/zenodo.24028384); v1.7, https://doi.org/10.5281/zenodo.24011489; toy study motivated by arXiv:2610.38932
Zenodo (this preregistration, PDF): https://doi.org/10.5281/zenodo.24048262
GitHub: https://github.com/wspaik5/ALP_ttbar_Toy_MC_v1.9_Preregistration_Layer_Geometry
Status: plan only. No v1.9 outcome exists at the time of this commit.
PDF version: same text plus two figures drawn only from the geometry definitions, an analytic resolution formula and published v1.6/v1.8 numbers (no new simulation or fit).

0. Why v1.9

v1.8 found that, in this toy, the 17 MeV loss is set mainly by multiple scattering relative to the small opening angle (median 0.42 mrad), not by the bremsstrahlung-reduced momentum: a diagnostic covariance with the true momentum at every surface removed most conversion fakes (7.4% → 1.9%) but improved 17 MeV only from 117× to 113×. Reducing the material helped (z at every layer: 117× → 56× → 26.7× for material scale factors f = 1, 0.5, 0.25). The v1.8 future directions named the layer geometry as the next test: a first layer closer to the beam, and more, thinner layers.

v1.9 asks whether **where** the material and the measurements are placed matters, at **fixed total material**.

An analytic estimate made before this preregistration (Fig. P1a; two 10 GeV tracks, z at every layer, Highland scattering, Fisher matrix of the r–z fit, no simulation) gives the r–z opening resolution σ(Δcotθ) at f = 0.25:

vertex radius	G0 (v1.8)	G1 closer	G2 finer	G3 both
1 mm	0.161 mrad	0.165	0.134	0.138
40 mm	0.163	0.163	0.155	0.156
100 mm	0.193	0.193	0.177	0.177
250 mm	0.203	0.203	0.203	0.203

For comparison: G0 at f = 1 gives 0.265–0.381 mrad, and the hit resolution alone 0.016–0.023 mrad. Scattering dominates by an order of magnitude, so at fixed material the geometry changes the resolution by at most about 17% (G2), and the closer first layer (G1) by almost nothing. On the other hand, more pixel layers mean more material bands, and the material veto removes signal pairs whose vertex lies near any band. v1.9 tests whether these two effects leave the conversion-induced loss essentially unchanged, as the estimate suggests.

0a. Related work (prior art)

- Low-material trackers with many thin layers and a first layer very close to the beam are pursued by next-generation concepts such as ALICE 3 [D1], ALICE ITS3 [D2] and the FCC-ee detector concepts [D3]. The geometries below only change the layer radii in this toy and make no statement about these detectors.
- The dependence of track and vertex resolution on layer radii, material and lever arm is textbook tracking [D4].

What v1.9 adds is limited to this series: it quantifies, in the preregistered toy of v1.0–v1.8, how the layer geometry at fixed material affects the conversion-induced loss for top-associated displaced ALP → e^+e^-. No claim of novelty for the method is made.

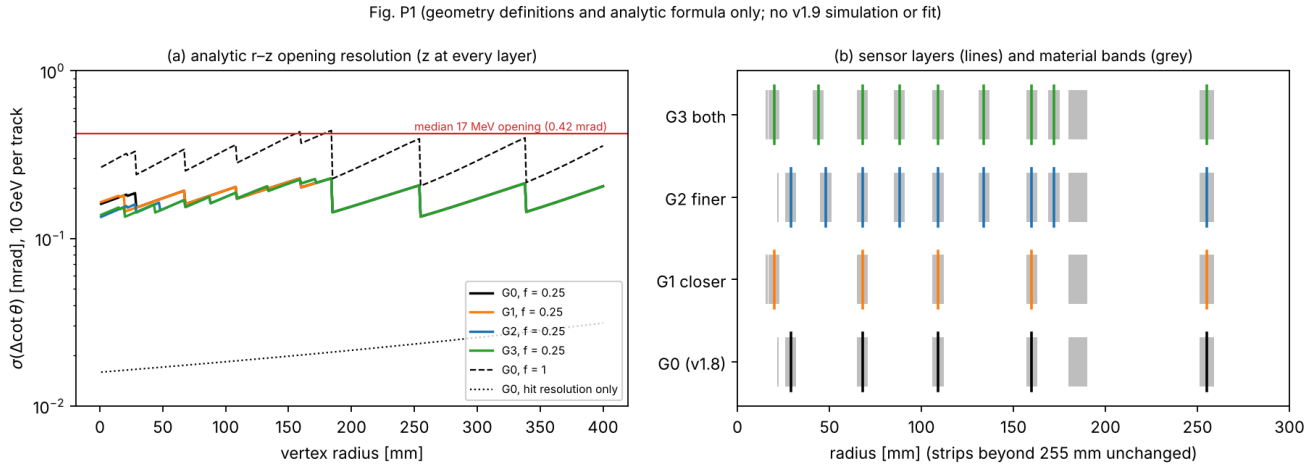


Fig. P1 (geometry definitions and an analytic formula only; no simulation). (a) Analytic r–z opening resolution σ(Δcotθ) for two 10 GeV tracks from a common vertex, z at every layer, Highland scattering at every material surface after the vertex (Fisher matrix of the r–z fit; no bremsstrahlung), for G0–G3 at f = 0.25, G0 at f = 1 and G0 without scattering; the red line is the median true polar opening of 17 MeV ALP pairs (v1.7). (b) Sensor layers and material bands of the four geometries.

1. Questions

- **Q1 (primary).** At fixed total material (f = 0.25, 3.8 T) and with z at every layer, how does the limit degradation from conversions at 17, 50, 100 and 200 MeV ($c\tau$ = 1 mm) change when the first layer is moved closer to the beam, when the pixel layers are doubled at half the thickness each, and both, compared with the v1.8 geometry (26.7×, 8.8×, 3.3×, 1.1×)?
- **Q2 (mechanism).** Do more, thinner layers improve the r–z separation (fraction of ALP pairs with $\Delta\chi^2_L > 9$) as estimated, and how much signal does the larger number of material bands cost in the material veto?

2. Unchanged from v1.8 (frozen)

- Field 3.8 T; material scale factor f = 0.25 applied to every material surface; v1.6 scalings of the conversion yield and the mass resolution for f = 0.25.
- Signal samples, kinematics, seeds and base selection (the base selection does not depend on the layers); subsample of the first 100,000 base conversion pairs and the first 60,000 base signal pairs per mass.
- Mini tracking simulation and fits (v1.3–v1.5), longitudinal simulation and r–z fit of v1.7 with both r–z hypotheses using s from the H_conv transverse fit, z at every layer with σ_z = 25 μm (V-allpixel), r–φ resolution 15 μm for layers below 200 mm and 35 μm above, nominal covariance (true initial momentum). Seeds 31/41 (transverse) and 1031/1041 (longitudinal).
- $\Delta\chi^2_3D = \Delta\chi^2_T + \Delta\chi^2_L$; selection `scan_v15` ($k \in \{\text{none}, 1, 2, 4, 9, 16, 25, 49\}$), material veto, counting rule and metrics as in v1.8.

3. What is new (frozen)

3.1 Geometries (radii in mm; strips at 255–1080 mm and all other material unchanged; Fig. P1b):

geometry	pixel layers (z and r-φ)	beam pipe	thickness per pixel layer
G0 (= v1.8)	29, 68, 109, 160	22	0.02 X_0 × f
G1 closer	20, 68, 109, 160	16	0.02 X_0 × f
G2 finer	29, 48, 68, 88, 109, 134, 160, 172	22	0.01 X_0 × f
G3 both	20, 44, 68, 88, 109, 134, 160, 172	16	0.01 X_0 × f

The total pixel material (0.08 X_0 × f), the beam-pipe thickness (0.003 X_0 × f) and all other material are identical in all geometries. Each pixel layer is a sensor and a material band of half-width 3 mm (beam pipe 0.5 mm), as before.

3.2 Samples. The conversion sample is regenerated for each geometry with the same generator and seed (5), because the conversion radii follow the material; the signal truth samples are unchanged. The geometry is selected by an environment variable before import, as the field and material were in v1.6.

3.3 Reporting. For each geometry: $\Delta\chi^2_T$ only and $\Delta\chi^2_3D$ (V-allpixel), with and without mass window; fraction of pairs with $\Delta\chi^2_L > 9$ for conversions (r < 800 mm) and each ALP mass; fraction of base signal pairs removed by the material veto (n_σ = 2, without exemption).

4. Sanity gate (frozen)

Before any degradation is computed:

1. **G0 identity.** Transverse vertex radius, $\Delta\chi^2_T$ and $\Delta\chi^2_L$ (V-allpixel) equal v1.8 config M025 within 10^{-6} for ≥ 99.9% of pairs in every sample.
2. **Material bookkeeping.** The total material (ΣX_0 at $\eta = 0$) is identical in G0–G3 to 10^{-12}.
3. **Fit success** ≥ 95%; fraction with $\Delta\chi^2_L < -10^{-6}$ ≤ 0.1% in every sample.

If the gate fails, the code is fixed, the failure and the fix are reported, and the gate is re-run.

5. Validation (independent samples only; criteria frozen)

- **U1 (null, G2).** The v1.3 V4 collinear sample (seed 404, longitudinal seed 1404), no bremsstrahlung, G2, V-allpixel: $\Delta\chi^2_L$ median 0.30–0.65 and 3–8% above 3.84 (χ^2_1).
- **U2 (propagation, G3).** Without scattering, z hits agree with $z_v + \cot\theta \cdot s$ to better than 0.1 µm.
- **U3 (response).** Two 10 GeV tracks from a vertex at r uniform in 30–100 mm, $|\eta| < 1$, polar opening 1 mrad, no bremsstrahlung (seed 808, longitudinal seed 1808): the median $\Delta\chi^2_L$ is larger in G2 than in G0.

If a criterion fails, the code is fixed and the fix is reported.

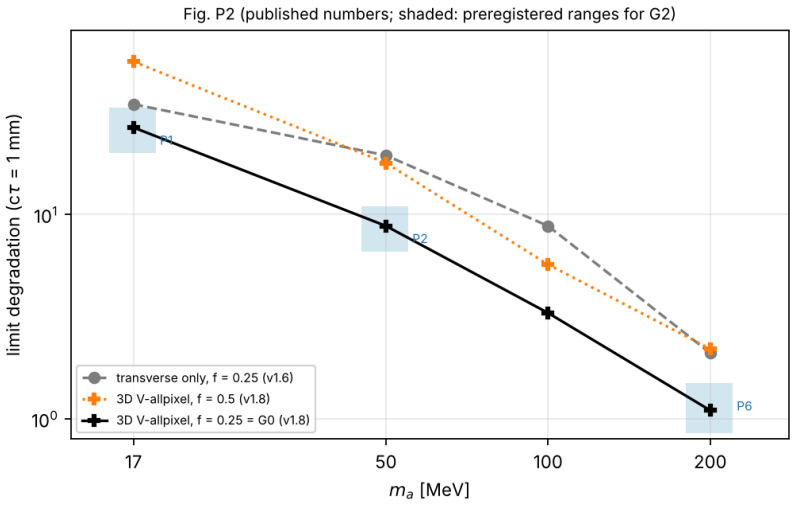


Fig. P2 (published v1.6/v1.8 numbers only; no v1.9 simulation or fit). Published degradations at f = 0.25 and 0.5 and the preregistered ranges of P1, P2 and P6 for G2 (shaded).

6. Predictions (written before running)

Values are degradations with the $\Delta\chi^2_3D$ (V-allpixel) selection, f = 0.25, cτ = 1 mm, compared with G0 (v1.8: 26.7×, 8.8×, 3.3×, 1.1×).

- **P1 (17 MeV, primary).** G2 is within ±25% of G0: **between 20× and 33.4×**. The geometry at fixed material does not give a gain comparable to the material reduction (v1.8: a factor 2.1 from f = 0.5 to 0.25).
- **P2 (50 MeV, primary).** G2 is **between 6.6× and 11×** (±25% of 8.8×).
- **P3 (closer first layer).** G1 is within ±10% of G0 at every mass.
- **P4 (combination).** G3 is within ±10% of G2 at every mass.
- **P5 (r–z mechanism).** The fraction of 17 MeV ALP pairs with $\Delta\chi^2_L > 9$ is higher in G2 than in G0 (61.4%), by **at most 10 percentage points**; the conversion fraction with $\Delta\chi^2_L > 9$ changes by at most a factor 1.5 from G0 (2.03%).
- **P6 (200 MeV).** All geometries give **≤ 1.5×**.
- **P7 (veto cost).** The fraction of base 17 MeV signal pairs removed by the material veto is **higher in G2 than in G0**.

Each prediction is reported as confirmed or refuted against these statements. Refuted predictions are not re-framed.

7. Reporting

v1.9 note with a Figure 13: (a) degradation versus mass for G0–G3 with the preregistered ranges; (b) the fraction of pairs with $\Delta\chi^2_L > 9$ for conversions and ALP pairs and the veto loss per geometry; (c) the analytic estimate of Fig. P1a next to the simulated r–z separation. The note contains the U1–U3 validation, the prediction table, "Deviations from preregistration", "Limitations" and "Future directions".

8. Limitations known in advance

- All v1.0–v1.8 limitations apply (barrel only, idealised cylindrical layers, no pattern recognition, toy material map, crude counting rule, selection optimised on the same samples, truth used for the material list and the scattering momentum, single-Gaussian bremsstrahlung model in the transverse fit, sequential 3D fit).

- Only the pixel region changes; real trackers also change supports, services and the strip region. Thinner layers are assumed to keep the same hit resolution.
- The analytic estimate of Sec. 0 ignores bremsstrahlung, track curvature and the vertex spread; it guides the predictions but is not a result.
- Subsampled statistics; only trends and ratios are meaningful. No statement about real detectors is made.

[D1] ALICE Collaboration, Letter of intent for ALICE 3, arXiv:2211.02491.
[D2] ALICE Collaboration, Technical Design Report for the ALICE Inner Tracking System 3 – ITS3, CERN-LHCC-2024-003.
[D3] N. Bacchetta et al., CLD – A Detector Concept for the FCC-ee, arXiv:1911.12230; The IDEA detector concept for FCC-ee, arXiv:2502.21223.
[D4] R. Frühwirth, M. Regler, R. K. Bock, H. Grote, D. Notz, Data Analysis Techniques for High-Energy Physics, 2nd ed., Cambridge University Press (2000).